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1 Linear algebra

Definition 1.1. Matrix over a set S.

"A is a matrix over a set S" $\longleftrightarrow \exists m, n \in \mathbb{N} : \{1, \ldots, m\} \times \{1, \ldots, n\} \longrightarrow S$

Definition 1.2.

$$\text{combine}(f_1, f_2) := \{(((m,k),(m,v)),(m,(k,v)))|(m,k) \in f_1 \wedge (m,v) \in f_2\}$$

(A combination is a product of morphisms in the category sets and functions.)

2 Topology

Definition 2.1.

$$\text{Segm} = \{[-\infty;\beta) : \beta \in \mathbb{R}\} \cup \{(\alpha;+\infty] : \alpha \in \mathbb{R}\} \cup \{(\alpha;\beta) : \alpha \in \mathbb{R}, \beta \in \mathbb{R}\}$$

Definition 2.2.

$$\mathcal{T}_{[-\infty;+\infty]} = \text{Segm}_\sigma$$

Definition 2.3. Set of neighbourhoods

$$\mathcal{N}_\mathcal{T}(x) = \{S \subseteq \bigcup \mathcal{T} : \exists U \in \mathcal{T} : x \in U \subseteq S\}$$

Definition 2.4. Interior of a set V in a topology $\mathcal{T}$.

$$\text{int}_\mathcal{T}(V) = \bigcup \{W \in \mathcal{T} : W \subseteq V\}$$

Theorem 2.1.

$$(\text{int} = \mathcal{N}^{-1}) \ \ \& \ \ (\mathcal{N} = \text{int}^{-1})$$

Theorem 2.2.

$$V \in \mathcal{N}_\mathcal{T}\langle A \rangle \longleftrightarrow A \subseteq \text{int}(V)$$

Theorem 2.3.

$$\text{cl} \dashv \text{id} \dashv \text{int}$$

Theorem 2.4.

$$V \in \mathcal{N}_\mathcal{T}(x) \longleftrightarrow x \in \text{int}_\mathcal{T}(V)$$

Proof.
($\Rightarrow$)
$V \in \mathcal{N}_{\mathcal{T}}(x) \implies \exists U \in \mathcal{T} : x \in U \subseteq V$ (nghb-elim, union-elim).
$U \in \mathcal{T} \ \& \ U \subseteq V \implies U \in \{W \in \mathcal{T} : W \subseteq V\}$ (specification-intro).
$x \in U \ \& \ U \in \{W \in \mathcal{T} : W \subseteq V\} \implies x \in \bigcup\{W \in \mathcal{T} : W \subseteq V\}$ (union-intro).
$x \in \bigcup\{W \in \mathcal{T} : W \subseteq V\} \implies x \in \text{int}_{\mathcal{T}}(V)$ (interior-intro).
($\Leftarrow$)
$x \in \text{int}_{\mathcal{T}}(V) \implies \exists W \in \{U \in \mathcal{T} : U \subseteq V\} : \ x \in W$ (interior-elim, union-elim).
$W \in \{U \in \mathcal{T} : U \subseteq V\} \implies W \in \mathcal{T} \ \& \ W \subseteq V$ (specification-elim).
$x \in W \ \& \ W \in \mathcal{T} \ \& \ W \subseteq V \implies \exists U \in \mathcal{T} : x \in U \subseteq V$ (and-intro, exists-intro).
$(\exists U \in \mathcal{T} : x \in U \subseteq V) \implies V \in \mathcal{N}_{\mathcal{T}}(x)$ (exists-elim, nghb-intro).
Done.

3 Measure theory

4 Functional analysis

5 Probability theory

6 Category theory